INCH-POUND
MIL-DTL-16377/86B(SH)
2 November 2021
SUPERSEDING
MIL-DTL-16377/86A(SH)
24 February 2015

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; LAMP, SOLID-STATE, FOR USE AS REPLACEMENT FOR COMMERCIAL FLUORESCENT LAMP

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

Unless otherwise noted, all dimensions are in inches. For tolerances other than those specified herein, see 3.5.2 of MIL-DTL-16377.

Dimensions provided on figures 1 through 6 are nominal dimensions meant to simplify identifying lamps. Actual required dimensions are discussed later in the document.

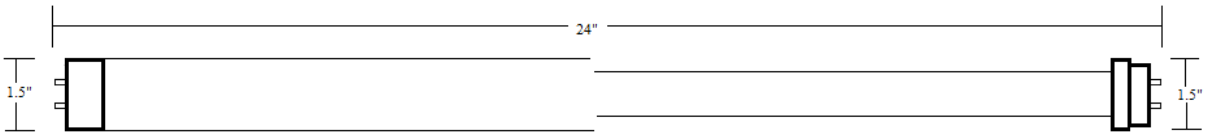


FIGURE 1. M16377/86-001 lamp dimensions.

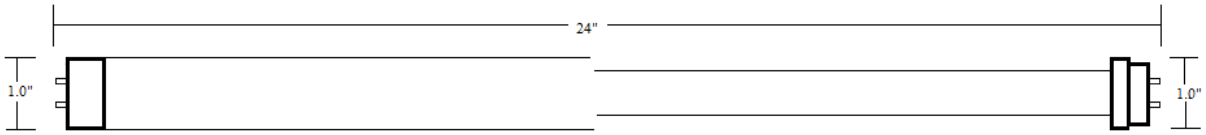


FIGURE 2. M16377/86-002 lamp dimensions.

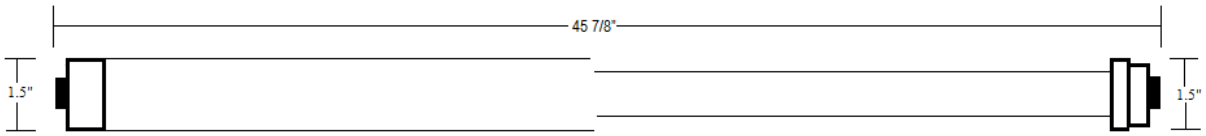


FIGURE 3. M16377/86-003 lamp dimensions.

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

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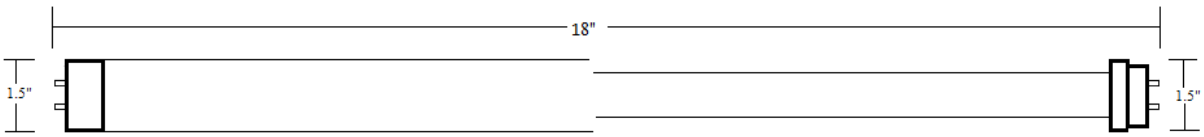


FIGURE 4. M16377/86-004 lamp dimensions.

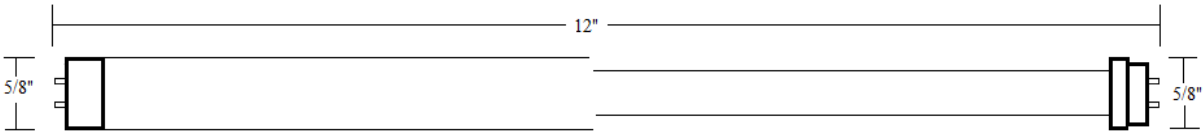
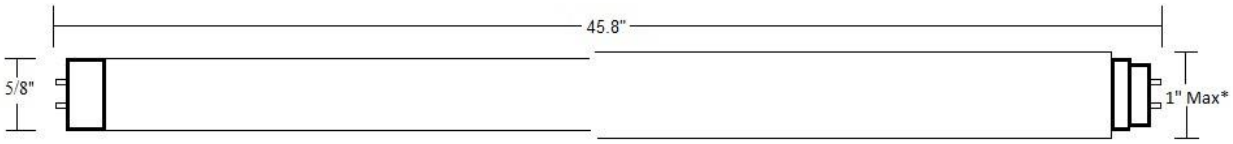


FIGURE 5. M16377/86-005 lamp dimensions.



* Lamp width wider than 5/8" allowable only when physical testing (shock, vibration) is succesfully passed in M16377/85 fixture with support clips removed. 1" maximum width allowed to ensure proper light output of LED arrays for color illumination.

FIGURE 6. M16377/86-006 lamp dimensions.

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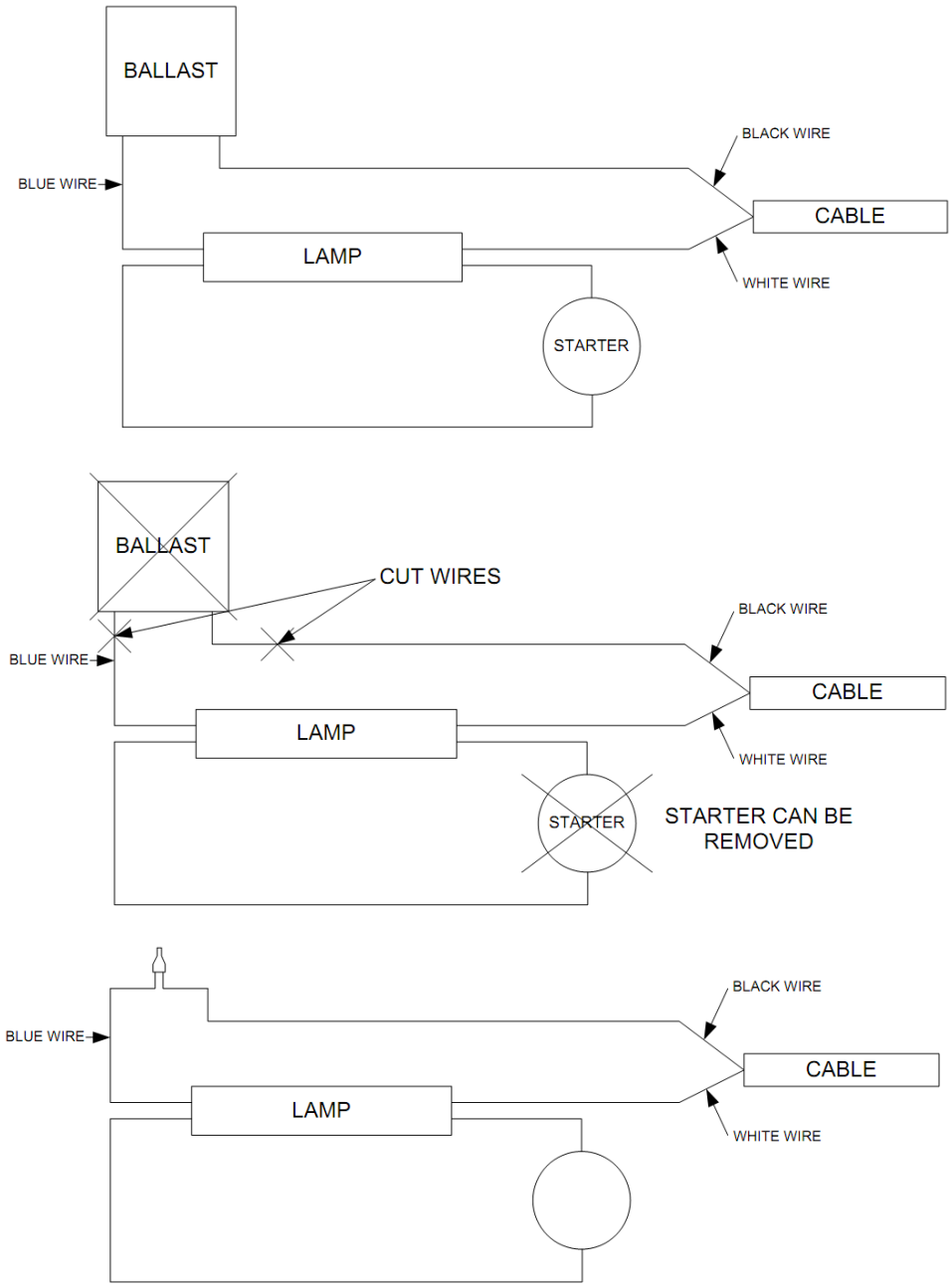


FIGURE 7. Ballast and starter removal for fixtures utilizing M16377/44, 2-wire ballast for 115-VAC operation of solid-state lamp.

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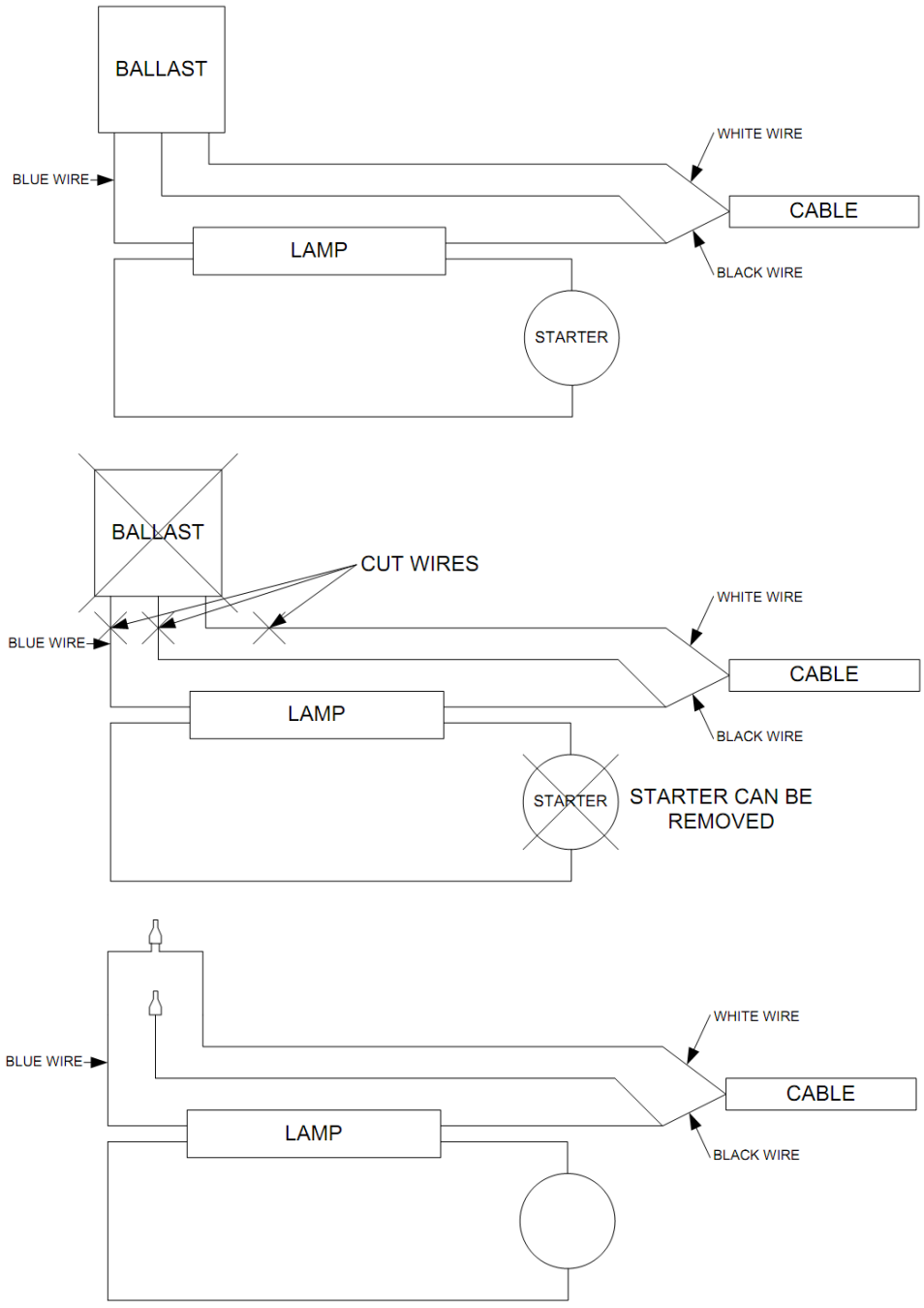


FIGURE 8. Ballast and starter removal for fixtures utilizing M16377/72, 3-wire ballast and M16377/81-002 ballast for 115-VAC operation of solid-state lamp.

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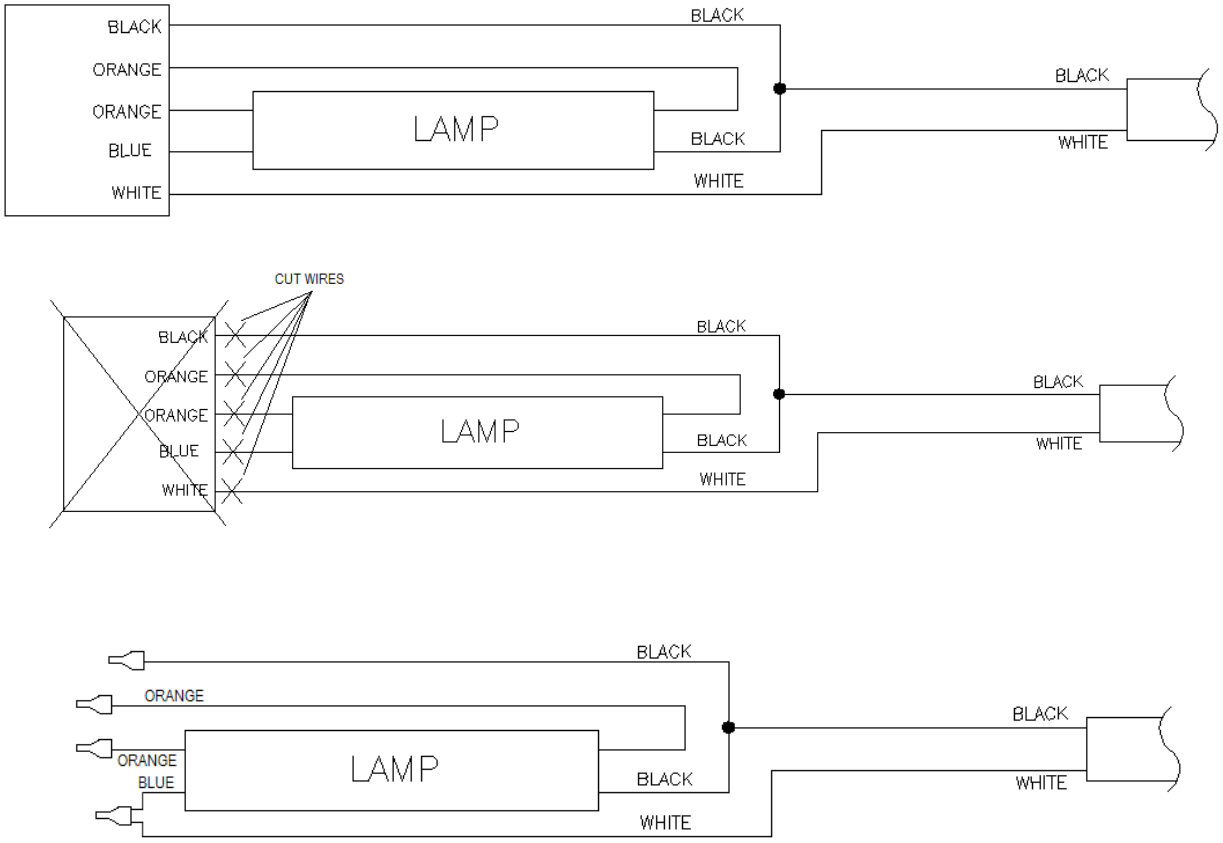


FIGURE 9. Ballast removal for fixtures utilizing M16377/81-001, 5-wire ballast for 115-VAC operation of solid-state lamp.

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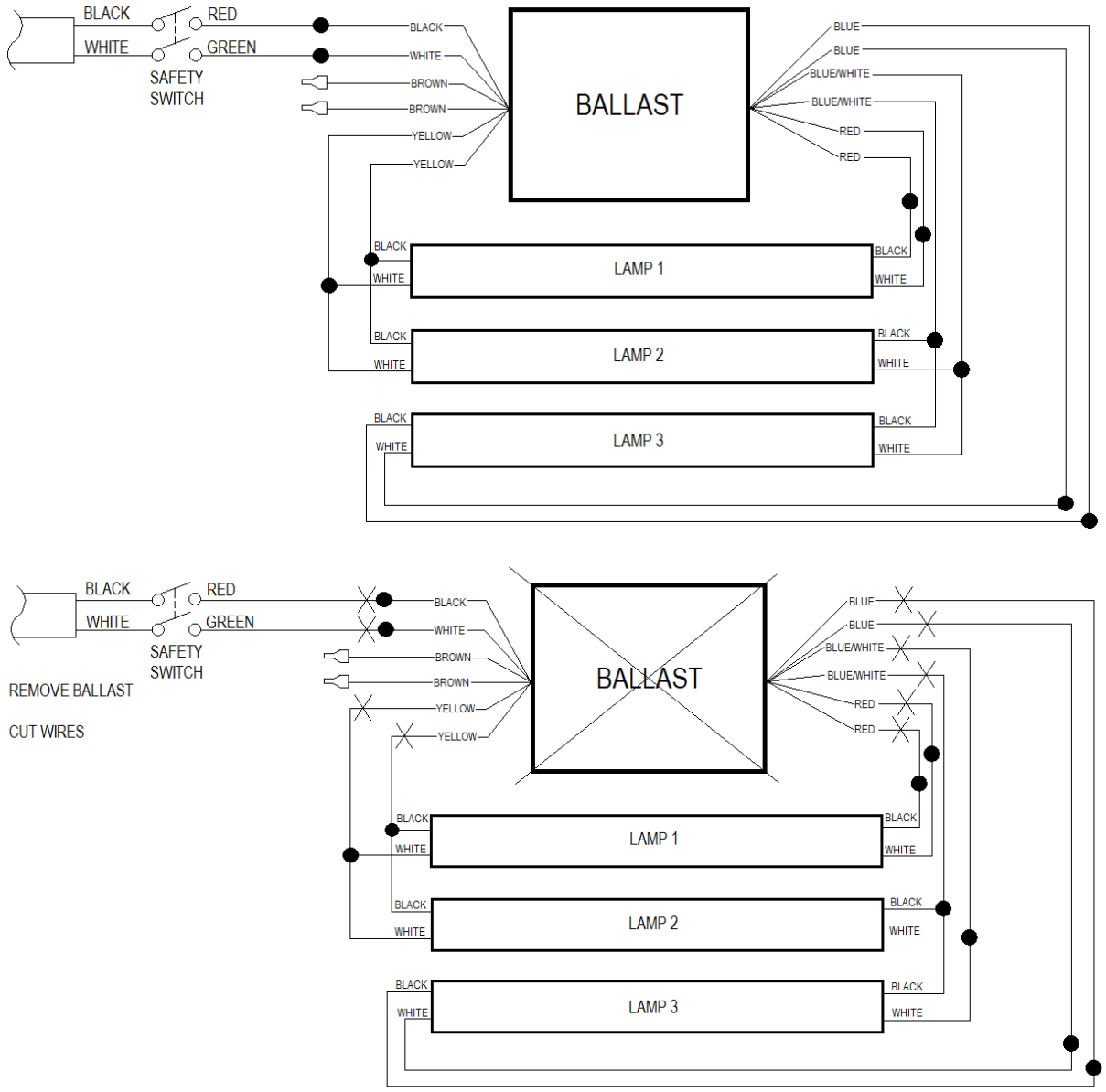


FIGURE 10. Ballast removal for M16377/13 fixtures for operation of solid-state lamp.

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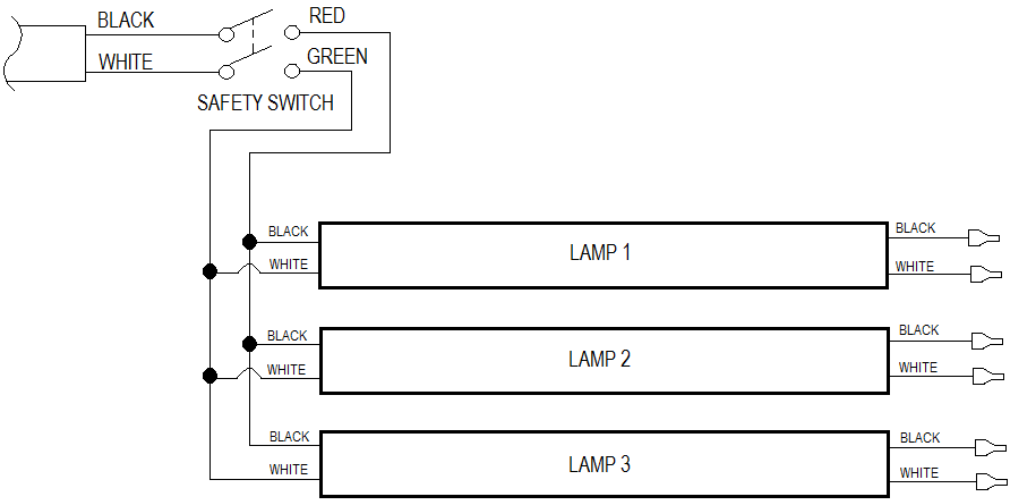


FIGURE 10. Ballast removal for M16377/13 fixtures for operation of solid-state lamp – Continued.

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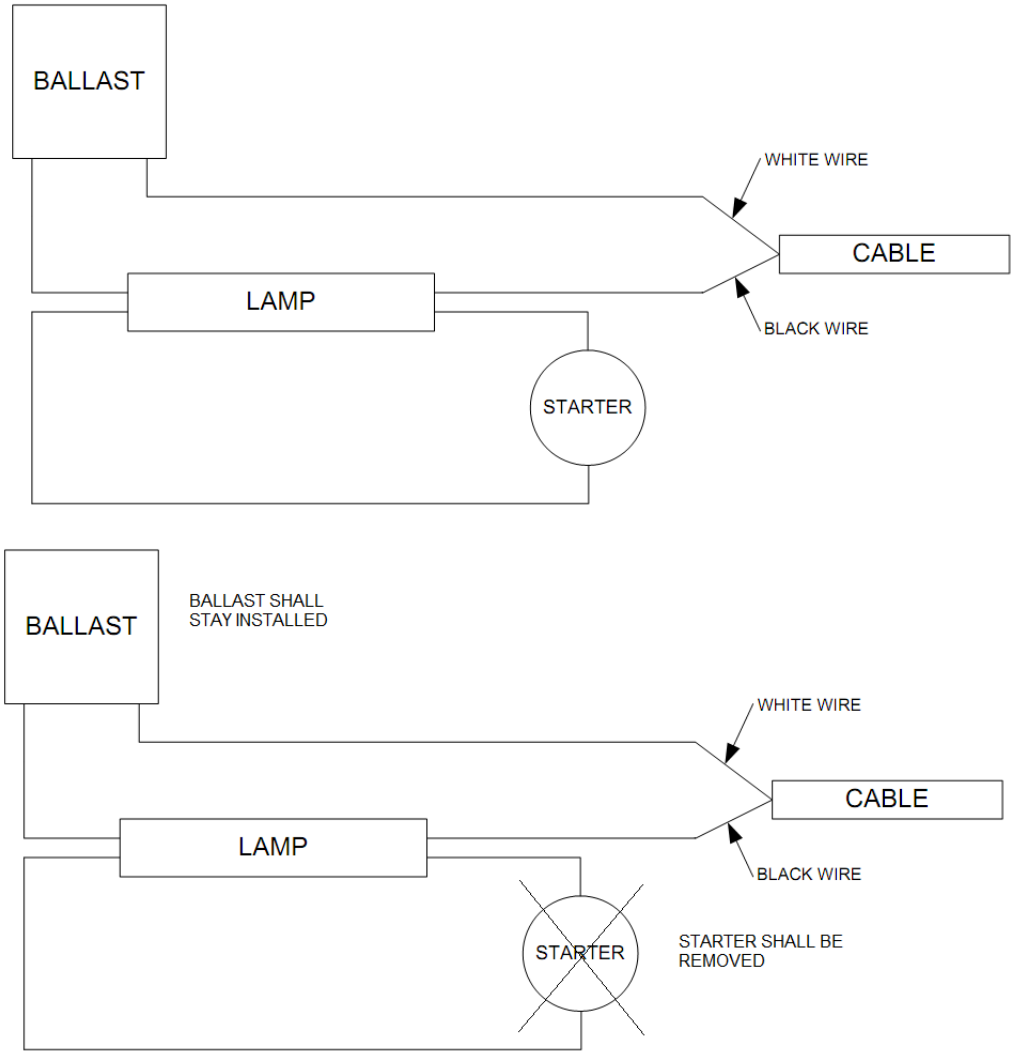
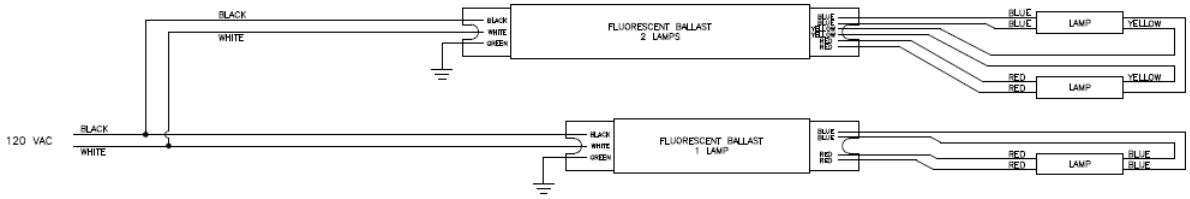
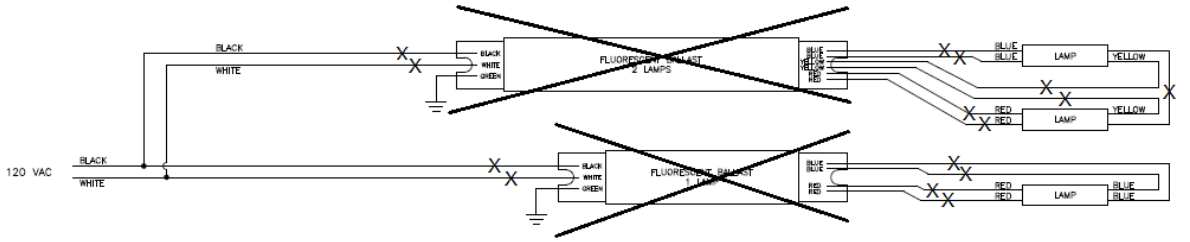


FIGURE 11. Starter removal for fixtures utilizing M16377/45 ballasts.

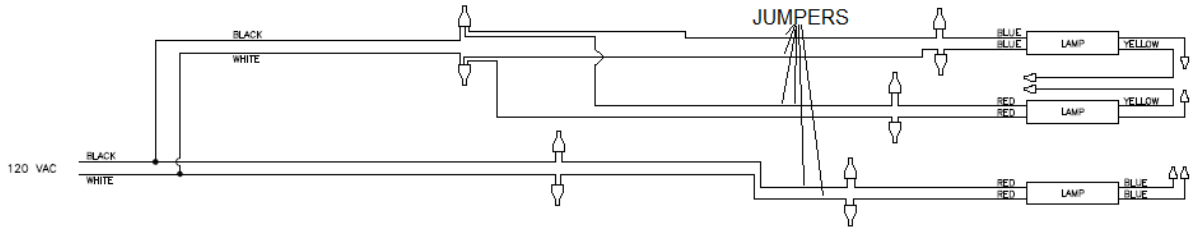
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Note: Only fluorescent circuit shown in figure. LED circuit removed from diagram for clarity.
Refer to Figures 4 and 5 of MIL-DTL-16377/85.



Note: Some versions of M16377/85 fixtures contain EMI filters prior to the inputs to the fluorescent ballasts. Testing shall be completed on fixtures with all filters removed.



Jumpers shall be 18 AWG, stranded, 105 degrees Celsius rated wire

FIGURE 12. Ballast removal for M16377/85 fixtures for operation of solid-state lamp.

P/N: M16377/86-XXX

NSN: 6240-XX-XXXX

115 VAC, 60 HERTZ, XX WATTS

FIGURE 13. Label for lamp; dimensions at manufacturer's discretion.

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REQUIREMENTS

CLASSIFICATION

Type: III
Class: 2

DESIGN

Electrical rating:

M16377/86-001, -002:	12 watts, maximum
M16377/86-003:	35 watts, maximum
M16377/86-004:	8 watts, maximum
M16377/86-005:	5 watts, maximum
M16377/86-006:	45 watts, maximum

Dimensions: The dimensions for the lamps are based on the commercial ANSI, NEMA, and IEC requirements for fluorescent lamps to the greatest extent possible. Changes from allowed tolerances in the commercial specifications may be necessary to ensure proper interfacing with existing lamp holders, lamp locks, or colored filter tubes in use on US Navy vessels. The internal lamp diameter between end caps may be less than or greater than the end cap diameter, as shown on figures 1 through 6.

End-cap length: The length of the end cap from the end of the pins or recessed socket to the light emitting portion of the lamp is at manufacturer's discretion, provided the lamp properly engages with existing lamp holders, lamp locks, and filter tubes as required.

M16377/86-001:	Nominal end-cap diameter is 1.5 inches. Measured diameter shall be within 1.41 and 1.52 inches. Length of the lamp from end of base pin to end of opposite pin shall be between 23.67 and 23.78 inches.
M16377/86-002:	Nominal end-cap diameter is 1.0 inch. Measured diameter shall be within 0.940 to 1.047 inches. Length of the lamp from end of base pin to end of opposite pin shall be between 23.67 and 23.78 inches.
M16377/86-003:	Nominal end-cap diameter is 1.5 inches. Measured diameter shall be within 1.41 and 1.59 inches. Length of the lamp from end of opposite base bosses shall be between 45.72 and 45.91 inches.
M16377/86-004:	Nominal end-cap diameter is 1.5 inches. Measured diameter shall be within 1.41 and 1.52 inches. Length of the lamp from end of base pin to end of opposite pin shall be between 17.67 and 17.78 inches.
M16377/86-005:	Nominal end-cap diameter is 0.625 (⅝) inch. Measured diameter shall be within 0.53 and 0.63 inch. Length of the lamp from end of base pin to end of opposite pin shall be between 11.72 and 11.91 inches.
M16377/86-006:	Nominal end-cap diameter is 0.625 (5/8) inch. Measured diameter shall be within 0.53 and 0.67 inch. Length of lamp from end of base pin to end of opposite pin shall be between 45.62 and 45.80 inches.

Mechanical: The size and shape of the light emitting diode (LED) lamp shall ensure that all existing lamp locks and shock clips within the legacy fixture fully engage when the LED lamp is installed. An exception to this requirement is allowed for M16377/86-006 LED lamps only when mechanical shock and vibration testing (see table I) is successfully passed in an M16377/85 fixture with the shock clips removed.

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Operation:

M16377/86-001: The solid-state lamp shall utilize LEDs and is intended as a replacement of the F20T12 and F17T12 fluorescent lamps used in the following MIL-DTL-16377 fixtures:

M16377/8
M16377/11
M16377/12
M16377/57
M16377/73
M16377/74
M16377/75

M16377/86-002: The solid-state lamp shall utilize LEDs and is intended as a replacement of the F17T8 fluorescent lamps used in the following MIL-DTL-16377 fixtures:

M16377/78
M16377/79
M16377/80
M16377/82
M16377/83
M16377/84

M16377/86-003: The solid-state lamp shall utilize LEDs and is intended as a replacement of the F48T12HO fluorescent lamps used in the following MIL-DTL-16377 fixture:

M16377/13

M16377/86-004: The solid-state lamp shall utilize LEDs and is intended as a replacement of the F15T12 fluorescent lamps used in the following MIL-DTL-16377 fixtures:

M16377/5
M16377/18
M16377/58
M16377/65
M16377/66

M16377/86-005: The solid-state lamp shall utilize LEDs and is intended as a replacement of the F8T5 fluorescent lamps used in the following MIL-DTL-16377 fixtures:

M16377/7
M16377/9
M16377/14
M16377/16
M16377/17
M16377/49
M16377/59

M16377/86-006: The solid-state lamp shall utilize LEDs and is intended as a replacement of the F54T5HO fluorescent lamps used in the following MIL-DTL-16377 fixture:

M16377/85

Electrical operation:

M16377/86-001: The lamp shall be compatible with M16377 fixtures utilizing F20T12 or F17T12 fluorescent lamps and the following ballasts, with the starter removed:

M16377/44
M16377/72

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The lamp shall also function in M16377 fixtures utilizing F20T12 or F17T12 fluorescent lamps when the starter has been removed, the ballast has been removed in accordance with figure 7 or 8, and ungrounded 115 volts alternating current (VAC) is applied to the lamp holders.

The lamp shall function in New Emergency Automatic Lighting Systems (NEALS) fixtures on battery power for a minimum of 5 hours utilizing a battery backup ballast that meets the requirements of MIL-DTL-16377/77.

M16377/86-002: The lamp shall be compatible with M16377 fixtures utilizing the M16377/81-001 ballast with no fixture modification. The lamp shall be compatible with M16377 fixtures utilizing the M16377/81-002 ballast with the starter removed.

The lamp shall also function in M16377 fixtures when the fixture has been modified as on figure 8 or 9 and ungrounded 115 VAC is applied to the lamp holders.

The lamp shall function in NEALS fixtures on battery power for a minimum of 5 hours utilizing a battery backup ballast that meets the requirements of MIL-DTL-16377/77.

M16377/86-003: The lamp shall function in M16377/13 fixtures when the ballast is removed in accordance with figure 10 and with ungrounded 115 VAC applied to the lamp holders.

M16377/86-004: The lamp shall be compatible with M16377 fixtures utilizing F15T12 fluorescent lamps and the following ballasts, with the starter removed:

M16377/44
M16377/72

The lamp shall also function in M16377 fixtures utilizing F15T12 fluorescent lamps when the starter has been removed, the ballast has been removed in accordance with figure 7 or 8, and ungrounded 115 VAC is applied to the lamp holders.

M16377/86-005: The lamp shall be compatible with M16377 fixtures utilizing F8T5 fluorescent lamps and ballasts manufactured to MIL-F-16377/45. The lamp shall function with the ballast remaining installed, the starter removed in accordance with figure 11, and with ungrounded 115 VAC applied to the lamp holders.

M16377/86-006: The lamp shall function in M16377/85 fixtures with the ballast removed in accordance with figure 12 and ungrounded 115 VAC applied to the lamp holders.

White light output:

M16377/86-001: The candlepower distribution curve of a modified symbol 333.1 fixture with the LED lamps installed shall meet the requirements of 3.7.1 of MIL-DTL-16377 when compared to the candlepower distribution curve of MIL-DTL-16377/12.

M16377/86-002: The candlepower distribution curve of a modified symbol 333.1T8 fixture with the LED lamps installed shall meet the requirements of 3.7.1 of MIL-DTL-16377 when compared to the white diffusing window candela distribution curve of MIL-DTL-16377/80.

M16377/86-003: The candlepower distribution curve of a modified symbol 74 fixture with the LED lamps installed shall meet the requirements of 3.7.1 of MIL-DTL-16377 when compared to the candlepower distribution curve of MIL-DTL-16377/13 (for replacement purposes only).

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M16377/86-004:	The candlepower distribution curve of a modified symbol 82 fixture with the LED lamps installed shall meet the requirements of 3.7.1 of MIL-DTL-16377 when compared to the symbol 82 candlepower distribution curve of MIL-F-16377/66 (for replacement purposes only).
M16377/86-005:	The candlepower distribution curve of a modified symbol 73.3 fixture with LED lamps installed shall meet the requirements of 3.7.1 of MIL-DTL-16377 when compared to the symbol 73.3 candlepower distribution curve of MIL-DTL-16377/9.
M16377/86-006:	The candlepower distribution curve of a modified symbol 76.X fixture with LED lamps installed shall meet the requirements of 3.7.1 of MIL-DTL-16377 when compared to the white mode candela distribution curve of MIL-DTL-16377/85. NOTE: Fixtures with any of the symbol numbers in MIL-DTL-16377/85 may be used.
Non-white light:	Non-white light shall be accomplished by using existing filters manufactured to MIL-DTL-16377/42, where applicable. No white light leakage shall be evident when existing filter tubes are used with M16377/86-001, -002, -004,or -005.
Lamp holder configuration (letters are commercial designations):	
M16377/86-001, -002, -004:	Compatible with medium bi-pin lamp holder (G13/G13M) of the naval-use lamp holders section of UL 496.
M16377/86-003:	Compatible with recessed double-contact lamp holder (R17d) of the naval-use lamp holders section of UL 496.
M16377/86-005, -006:	Compatible with miniature bi-pin lamp holder (G5) of the naval-use lamp holders section of UL 496.
Weight:	
M16377/86-001, -002:	0.95 pound, maximum
M16377/86-003, -006:	2 pounds, maximum
M16377/86-004:	0.70 pound, maximum
M16377/86-005:	0.25 pound, maximum
Color rendition:	In accordance with 3.7.4.1.2 of MIL-DTL-16377.
Color temperature:	4,100 K (see 3.7.4.1.1 of MIL-DTL-16377)
Designed lumen maintenance:	L70 ≥50,000 hours (see 3.7.11 of MIL-DTL-16377)
Minimum lumens at 0 hours on battery backup for M16377/86-001, -002:	240 (see 3.7.1.1 of MIL-DTL-16377)
Minimum lumens at 5 hours on battery backup for M16377/86-001, -002:	144 (see 3.7.1.1 of MIL-DTL-16377)
Circuit design:	The lamp assembly shall incorporate integrated driver circuitry so that no external driver components are required.
Designation and marking:	One label as shown on figure 11.

EXAMINATION AND TESTS

First article inspection:	See table I.
Conformance:	
Conformance inspection:	See MIL-DTL-16377.
Comparison inspection:	Same as the first article inspection.

GENERAL INFORMATION

Assignment of military part numbers and general information is shown in table II.

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TABLE I. First article and comparison inspection for complete LED tube. 1/

Tests	Requirement	Test method	Remarks
Examination	3.13	4.5	2/
Operation	3.13	4.8.1	3/
Dielectric withstanding voltage	3.6.15	4.8.2	3/, 4/
Insulation resistance	3.6.16	4.8.3	3/, 4/
Shock	3.5.12	4.8.7	3/, 4/
Vibration	3.5.13	4.8.8	3/, 4/
Dielectric withstanding voltage	3.6.15	4.8.2	3/, 4/
Insulation resistance	3.6.16	4.8.3	3/, 4/
Noise	3.5.14	4.8.9	3/, 4/
Photometrics	3.7.1	4.8.6.1	3/
Brightness	3.7.5	4.8.6.4	3/
Chromaticity	3.7.4	4.8.6.5	3/
Correlated color temperature	3.7.4.1.1	4.8.6.6	3/
Lumen maintenance	3.7.11	4.8.20	3/
Fail-safe circuit	3.6.17	4.8.24	--
Plastics flammability and smoke	3.4.2.1	--	--
Optical uniformity	3.7.7.1	4.8.4	2/
Electromagnetic interference	3.6.14	4.8.16	3/, 5/, 7/
Power interface	3.6.18	4.8.21	3/
Power quality	3.6.18.1	4.8.22	3/
Emergency light output	3.7.1.1	4.8.6.8	6/
Battery discharge test	3.6.19.3	4.8.26	6/

NOTES:

1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377.

2/ An individual LED tube shall be examined for weight, dimensions, optical uniformity, and product defects.

3/ Testing shall be accomplished as follows. The legacy fixture shall be modified as shown on figure 7, 8, 9, 10, 11, or 12, as appropriate.

a. With three LED lamps installed in an M16377/12 symbol 333.1 fixture for M16377/86-001.
b. With three LED lamps installed in an M16377/80 symbol 333.1T8 fixture for M16377/86-002.
c. With three LED lamps installed in an M16377/13 symbol 74 fixture for M16377/86-003.
d. With three LED lamps installed in an M16377/66 symbol 82 fixture for M16377/86-004.
e. With three LED lamps installed in an M16377/9 symbol 73.3 fixture for M16377/86-005.
f. With three LED lamps installed in an M16377/85 fixture, any style, as appropriate for M16377/86-006.

4/ Testing shall be completed in the above order.

5/ Required tests are RE101, RE102, CE101, and CE102 only.

6/ Testing shall be accomplished with two LED lamps installed in an M16377/74 symbol 77.4EM fixture, or two LED lamps installed in an M16377/83 symbol 77.4T8EM fixture, as appropriate. The legacy fixture shall be modified as shown on figure 7, 8, or 9, as appropriate.

7/ M16377/86-003 and M16377/86-006 lamps shall be tested to the “exposed below deck” requirement for RE102 for surface ships, in accordance with MIL-STD-461.

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TABLE II. Military part numbers.

Military part number M16377/86-	Item name
001	2-foot T12 LED lamp, white
002	2-foot T8 LED lamp, white
003	4-foot T12HO LED lamp, white
004	18-inch T12 LED lamp, white
005	1-foot T5 LED lamp, white
006	4-foot T5HO LED lamp, white

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 6210-2021-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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